QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12271	05/12/2023	CC274723000017088F	04/12/2023	03/12/2025	1 of 1

MRROA No. : QC1242 **Date of Receipt** 04/12/2023 **Customer DC. No.** : OFFICE003/23-24 dt: 04/12/2023

Customer : M/s Nihva Technologies Pvt. Ltd.,
Address 187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location : In Lab
Condition of Item : OK
UUC* Details : Coat Thickness Gauge
Range : 0-1350 µm
Identification No. : DFT-01
Id No. taken from : As per challan
Standard Referred : IS 6012

Make : Mextech
L.C. : 0.1 µm / 1 µm
Sl. No. : N1061412
Accuracy : ± 1.5% of reading & 2 µm whichever is more

Calibration Method : QS/CPM/21

Test Purpose : To check the actual coat thickness gauge with pre calibrated Qrics foils.

Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No.	**Standard / Equipment**	**Id. No.**	**Cert. No.**	**NABL Cert. No.**	**Validity**
1.	Foils	QS/MM/09 (A to G)	QC/22-23/10851 dated 02/01/2023	CC-2747	01/01/2024

Observations : **Pre Calibration Status** : Worth calibrating
: **Post Calibration Status** : As follows reading average value of three reading,

For Ferrous

Sl. No.	Actual Foils Size µm	CTG Set to Actual Foils size.	Actual Reading on CTG in µm	Repeatability in µm	Error	Error in %
1	25	25	24.6	0.2	-0.4	-1.6
2	51	51	51.4	1.0	0.4	0.8
3	125	125	125.0	3.0	0.0	0.0
4	252	252	252.0	1.0	0.0	0.0
5	515	515	516.0	4.0	1.0	0.2
6	708	708	709.0	2.0	1.0	0.1

For Non- Ferrous

Sl. No.	Actual Foils Size µm	CTG Set to Actual Foils size.	Actual Reading on CTG in µm	Repeatability in µm	Error	Error in %
1	25	25	25.2	0.8	0.2	0.8
2	51	51	51.1	0.9	0.1	0.2
3	125	125	124.0	2.0	-1.0	-0.8
4	252	252	253.0	2.0	1.0	0.4
5	515	515	515.0	4.0	0.0	0.0
6	708	708	710.0	5.0	2.0	0.3

Note: The customer is advised to select the shim nearest to the value under measurement and set to the true value of shim & zero by using ± buttons on meter. Then measure the actual value on component.

Uncertainty of Measurement : ± 4.74 µm

(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45 %.)

*** END ***

Calibrated By N. M. Shinde
Technical Assistant

Reviewed
08/12/23

Approved By Ankita Bhuvad
Authorized Signatory

User discretion:

Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

P.T.O.

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12270	05/12/2023	CC274723000017087F	04/12/2023	03/12/2025	1 of 1

MRROA No. : QC1242 **Date of Receipt** 04/12/2023 **Customer DC. No.** : OFFICE003/23-24 dt: 04/12/2023

Customer Address : M/s Nihva Technologies Pvt. Ltd.,
187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location : In Lab
Condition of Item : OK
UUC* Details : Foils
Size of foils : 125 to 705 µm
Id. No. of foils : Nil
Id. No. taken from : NA
Calibration Method : QS/CPM/21

Make : Unknown
L.C. : NA
Sl. No. : As below

Test Purpose : To check the actual thickness of foils
Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No.	**Standard / Equipment**	**Id. No.**	**Cert. No.**	**NABL Cert. No.**	**Validity**
1.	Universal Length measuring machine LMM 300T	QS/MM/37	CGIL/2023/1125/I-001 dt: 25/11/2023	CC-2656	24/11/2025

Observations : **Pre Calibration Status** : Worth calibrating,
: **Post Calibration Status** : As Follows

Mechanical Calibration: Dimension (Basic Measuring Instruments, Gauges etc.)

Sl. No.	Sl. No.of Foils	Specified Foils Size in µm	Actual Foils Size in µm
1	CM 100	125 ± 1	126
2	CM 300	282 ± 3	283
3	CM 500	393 ± 5	389
4	CM 800	705 ± 8	707

Uncertainty of Measurement : ± 2.2 µm
(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45%.)

Note: *UUC =Unit Under Calibration

Reviewed
08/12/23

*** END ***

Calibrated By N. M. Shinde, Technical Assistant

Approved By Ankita Bhuvad, Authorized Signatory

User discretion:
Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.